

PROJECT SUMMARY

1. Project Title;

Food Delivery Operations Analysis – Food Delivery Performance and Delay Analysis

2. Business Problem

The project analyzes food delivery operations to identify issues related to **delivery delays, order cancellations, customer satisfaction, and restaurant/location performance**. The main objective is to determine which restaurants and locations require improvement, understand whether delivery delays are associated with lower customer ratings, and provide data-driven recommendations to improve delivery efficiency and customer experience.

The analysis addresses key business questions such as restaurant order volume, locations with higher delays, restaurant cancellation rates, customer ratings by delivery performance, and overall restaurant performance.

3. Dataset

The project uses the **Kaggle Restaurant Delivery Orders Dataset**.

<https://www.kaggle.com/datasets/kbrakssa/restaurant-delivery-orders-dataset>.

Important fields include Order ID, Order Time, Delivery Time, Delivery Duration, City, State, Restaurant, Cancellation Status, Cancellation Reason, Customer Rating, Latitude, Longitude, and Delivery Status.

4. Data Cleaning and Transformation

Python and Pandas were used to load, inspect, clean, transform, and analyze the dataset. The main cleaning activities included:

- ✓ Checking rows, columns, data types, missing values, and duplicates.
- ✓ Converting order and delivery time columns into appropriate datetime formats.
- ✓ Handling missing values and validating cancellation-related fields.
- ✓ Checking consistency between delivery timestamps and delivery duration.
- ✓ Creating the **Delivery_Status** field based on delivery duration:
 - **Fast:** ≤30 minutes
 - **Normal:** 31–45 minutes
 - **Delayed:** >45 minutes

BeautifulSoup was used to collect external city-related information from a public webpage as part of the web-scraping requirement.

5. Key Findings

1. Order and cancellation performance:

Out of **10,000 orders**, **8,961 were completed** and **1,039 were cancelled**, resulting in an overall cancellation rate of **10.39%**.

2. Delivery delays are a major issue:

Among completed orders, the average delivery time was approximately **49.37 minutes**, with a median of **47 minutes**. Completed delivery times ranged from **20 to 90 minutes**. Approximately **53.97% of completed orders were delayed**.

3. Delivery time and customer satisfaction:

Fast deliveries had an average customer rating of **4.75/5**, Normal deliveries **4.48/5**, while Delayed deliveries averaged only **3.36/5**. Delivery duration and customer rating showed a strong negative association of approximately **-0.90**, indicating that longer deliveries are strongly associated with lower customer satisfaction.

4. Peak hours create operational pressure:

Average delivery time increased significantly during lunch and dinner peaks. The highest average delivery times were around **12 PM (59.37 minutes)**, **1 PM (59.56 minutes)**, **6 PM (60.14 minutes)**, and **7 PM (59.30 minutes)**.

5. City and restaurant hotspots:

New York had the highest delayed share at approximately **61.0%**, while **San Antonio** had the highest cancellation rate at approximately **11.7%**. At the restaurant level, **Pasta Place** had the highest average delivery time at approximately **49.85 minutes** and the highest delayed rate at **60.7%**. **Bistro Belle** had the highest restaurant cancellation rate at approximately **11.7%**.

6. Business Recommendations

1. Improve peak-hour capacity:

Increase delivery-partner availability and restaurant preparation capacity during the **12–1 PM** and **6–7 PM** peak periods. Improve dispatch planning and monitor preparation queues to reduce peak-hour delays.

2. Reduce extreme delivery times:

Identify and investigate orders with the longest delivery durations. Focus on bottlenecks in food preparation, order assignment, dispatching, and delivery operations. Reducing extreme delays can improve overall customer satisfaction.

3. Prioritize high-risk locations and restaurants:

Conduct detailed operational reviews of **New York** and **San Antonio**, as well as **Pasta Place** and **Bistro Belle**. Monitor their delay and cancellation KPIs regularly and implement targeted improvements.

7. Technical Requirements Demonstrated

The project demonstrates a complete Data Analyst workflow using multiple technologies:

- **Python:** Loading and processing the dataset.
- **Pandas:** Data cleaning, transformation, validation, and analysis.
- **BeautifulSoup:** Collecting approximately 10–20 external city-related records from a public webpage.
- **SQL/PostgreSQL:** Completing all five required business queries covering restaurant orders, location delivery time, restaurant cancellation rate, customer ratings by delivery status, and best-performing restaurants.
- **Excel:** Creating Pivot Tables, formulas, conditional formatting, business analysis, and charts.
- **Power BI:** Creating a one-page interactive **Delivery Operations Performance Dashboard** with KPIs, charts, and slicers for Location, Restaurant, and Delivery Status.

The dashboard focuses on useful business information rather than unnecessary visualizations. Each chart is designed to answer a specific business question and support decision-making.

8. Final Objective

The project demonstrates a simplified real-world **Data Analyst workflow** by showing how different tools work together to transform raw data into business decisions. Python and Pandas are used for cleaning and transformation, BeautifulSoup for external data collection, SQL for business queries, Excel for quick analysis and reporting, and Power BI for interactive visualization.

Overall, the analysis shows that **delivery delays, particularly during peak hours, are a major operational concern and are strongly associated with lower customer ratings**. Therefore, improving peak-hour capacity, reducing extreme delivery times, and addressing high-delay and high-cancellation hotspots should be the primary business priorities.

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